

Name: []

Points: []

/20

Databasutveckling och lagring — Final Exam: RETAKE II

2025-03-06

Instructions

1. No computers, no phones, no notes
2. No talking with other people in the room
3. In case point 1 or 2 are broken, you get automatically **0 points** on the test.
4. Write your name on top of the page (do not fill in the *Points* box!)
5. For each question, you'll need to answer the question by crossing with an **X** the right answer (multiple choice questions) or write your answer in the empty lines below the question (open questions).
6. Once you're done:
 - Double-check your answers
 - Hand over your test to me
 - Leave the room

NB: While inside the classroom, rule 1 and 2 are still valid, even if you've already handed over your answers to me. If you want to discuss your answers with a classmate, or to check an answer on your phone, do that **outside the classroom**.

NB: Make sure to use the restroom, fill our water bottle, grab a coffee, etc... **before** we begin the exam, as you're not allowed to leave the classroom without handing over your answers.

Point system

There are **10 questions**, worth **2 points** each, for a total of maximum **20 points**.

There are no partial points

You either give the **complete, correct answer**, and get 2 points, or get 0 points.

Grading

Points	Grade
0 - 12	IG
13 - 19	G
20	VG

1 Normalization

Given the following schema of a SQLite database:

Listing 1: The complete schema for the DB

```
PRAGMA foreign_keys = ON;
```

```
CREATE TABLE IF NOT EXISTS customer (  
  customer_id INTEGER PRIMARY KEY,  
  full_name VARCHAR(100) NOT NULL,  
  email VARCHAR(100) UNIQUE NOT NULL,  
  postal_code VARCHAR(5) NOT NULL,  
  city VARCHAR(50) NOT NULL  
);
```

```
CREATE TABLE IF NOT EXISTS orders (  
  order_id INTEGER PRIMARY KEY,  
  order_date TEXT NOT NULL,  
  customer_id INTEGER NOT NULL,  
  FOREIGN KEY (customer_id) REFERENCES customer(customer_id)  
);
```

```
CREATE TABLE IF NOT EXISTS products (  
  product_id INTEGER PRIMARY KEY,  
  product_name TEXT NOT NULL,  
  category TEXT,  
  unit_price FLOAT NOT NULL  
);
```

```
CREATE TABLE IF NOT EXISTS order_items (  
  order_id INTEGER NOT NULL,  
  product_id INTEGER NOT NULL,  
  quantity INTEGER NOT NULL,  
  product_name TEXT NOT NULL,  
  PRIMARY KEY (order_id, product_id),  
  FOREIGN KEY (order_id) REFERENCES orders(order_id),  
  FOREIGN KEY (product_id) REFERENCES products(product_id)  
);
```

Point out and explain **one** normalization problem present in the schema.

They are missing many to many relations

- 2 Explain how you would solve the normalization problem you described above.

You don't need to write a SQL query, just explain how you would fix it.

by creating a junction table

- 3 In this course we've labbed with a database that contained the data for a music store. What was the name of the database, and which RDBMS did it use?

Name: _____

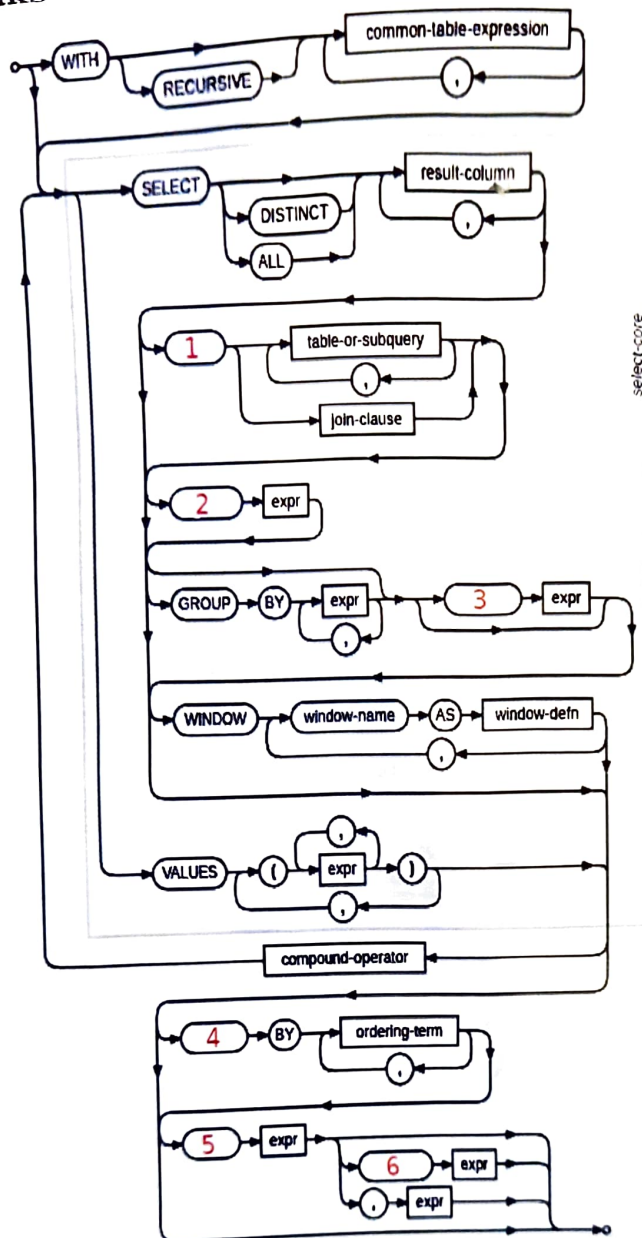
RDBMS: _____

- 4 In this course, we've labbed with a database that contained the data for a movie rental store. What was the name of the database, and which RDBMS did it use?

Name: _____

RDBMS: _____

5 Fill the blanks



1: select

2: update

3: update

4: ~~sort~~ filter

5: table

6: update

6 What's the difference between WHERE and HAVING in a SELECT query?

where filter by keyword while
having is never used

7 What's the difference between 'JOIN...USING' and 'JOIN...ON'?

8 We can always replace 'JOIN...ON' with 'JOIN...USING'

1. True

2. False

9 We can always replace 'JOIN...USING' with 'JOIN...ON'

1. True

2. False

10 What is a subquery?

an action that handles
results in the same
functions.